set_wifi

Command: `set_wifi`

Description

This API is used to set wifi configuration. It currently supports IPv4 addressing with DHCP-based IP assignment only; static IP configuration and IPv6 support is not available in the current API version.

Download pdf: [Download set_wifi as PDF](https://aa5123.github.io/zebra-rfd40-rfd90-iot-docs/command-pdfs/set_wifi.pdf)

Request

Request Fields

Field	Type	Required	Description	Example
`command`	string	Yes	Specifies the operation being performed, in this case, setting the wifi configuration.	`set_wifi`
`requestId`	string	Yes	A unique identifier for the request, allowing tracking and debugging of the operation.	`abc123`
`wifiConfig`	object	No		
`wifiConfig.interfaceName`	string	Yes	The name of the network interface for which the Wi-Fi configuration is being set.	
`wifiConfig.enableInterface`	boolean	Yes	Specifies whether the network interface should be enabled or disabled.	

Field	Type	Required	Description	Example
`wifiConfig.operation`	string (enum)	Yes	The operation to be performed on the Wi-Fi configuration. It can either create a new configuration or modify an existing one.	`CREATE`
`wifiConfig.accessPoint`	object	Yes		
`wifiConfig.accessPoint.connect`	boolean	Yes	Indicates whether to connect to this profile. If true, disconnects the currently connected profile and connects to the specified one; if false, just saves the profile.	
`wifiConfig.accessPoint.isPreferred`	boolean	Yes	Marks this profile as the preferred one to always connect to when available.	
`wifiConfig.accessPoint.autoConn`	boolean	No	Reserved for backward compatibility. It will be removed in the future.	
`wifiConfig.accessPoint.essid`	string	Yes	The ESSID (network name) of the access point.	`ZWireless`
`wifiConfig.accessPoint.enableSecurity`	boolean	Yes	Indicates whether security is enabled for this access point.	
`wifiConfig.accessPoint.ipv4Configuration`	object	No		
`wifiConfig.accessPoint.ipv4Configuration.ipAddress`	string	No	The IPv4 address to assign to the interface.	`192.168.1.81`
`wifiConfig.accessPoint.ipv4Configuration.subnetMask`	string	No	The IPv4 subnet mask.	`255.255.255.0`

Field	Type	Required	Description	Example
`wifiConfig.accessPoint.ipv4Configuration.gateway`	string	No	The IPv4 gateway address.	`192.168.1.144`
`wifiConfig.accessPoint.ipv4Configuration.enableDhcp`	boolean	No	Specifies whether DHCP is enabled for automatic IP address assignment.	
`wifiConfig.accessPoint.ipv4Configuration.dnsServer`	string	No	The IPv4 address of the DNS server.	`192.168.1.78`
`wifiConfig.accessPoint.ipv4Configuration.domainName`	string	No	The domain name for the network.	`example.test.com`
`wifiConfig.accessPoint.ipv6Configuration`	object	No		
`wifiConfig.accessPoint.ipv6Configuration.ipAddress`	string	No	The IPv6 address to assign to the interface.	`2a02:1234:420a:100b::99`
`wifiConfig.accessPoint.ipv6Configuration.prefix`	integer	No	The IPv6 address prefix length.	`65`
`wifiConfig.accessPoint.ipv6Configuration.gateway`	string	No	The IPv6 gateway address.	`192.168.1.144`
`wifiConfig.accessPoint.ipv6Configuration.enableAuto`	boolean	No	Specifies whether automatic IPv6 address assignment (e.g., SLAAC) is enabled.	
`wifiConfig.accessPoint.ipv6Configuration.dnsServer`	string	No	The IPv6 address of the DNS server.	`2001:4860:4860::9999`
`wifiConfig.accessPoint.ipv6Configuration.domainName`	string	No	The domain name for the IPv6 network.	`example.com`
`wifiConfig.accessPoint.security`	object	No		

Field	Type	Required	Description	Example
`wifiConfig.accessPoint.security.securityType`	string (enum)	Yes	The security type for the access point, specifying the authentication and encryption method.	`WPA2personal`
`wifiConfig.accessPoint.security.securityDetails`	object	Yes		
`wifiConfig.accessPoint.security.securityDetails.WPA2Personal`	object	No		
`wifiConfig.accessPoint.security.securityDetails.WPA2Personal.password`	string	Yes	The password for WPA2-Personal authentication.	`test@123`
`wifiConfig.accessPoint.security.securityDetails.WPA3Personal`	object	No		
`wifiConfig.accessPoint.security.securityDetails.WPA3Personal.password`	string	Yes	The password for WPA3-Personal authentication.	
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise`	object	No	WPA2-Enterprise security configuration with optional certificate parameters.	
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.authentication`	string (enum)	Yes	Authentication protocol for WPA2-Enterprise.	`peap`
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.innerAuthentication`	string (enum)	No	Inner authentication method to be used within the outer authentication protocol.	`tls`

Field	Type	Required	Description	Example
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.identity`	string	No	The identity (username) for WPA2-Enterprise authentication.	`test`
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.anonymousIdentity`	string	No	The anonymous identity for WPA2-Enterprise authentication.	`test`
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.password`	string	No	The password for WPA2-Enterprise authentication.	`test@123`
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.passphrase`	string	No	The passphrase for WPA2-Enterprise.	`test@123`
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.certificate`	array	No		`ap1wpa2e`
`wifiConfig.accessPoint.security.securityDetails.WPA2Enterprise.protocol`	string (enum)	No	The encryption protocol for WPA2-Enterprise.	`WPA2_Enterprise_CCMP`
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise`	object	No	WPA3-Enterprise security configuration with optional certificate parameters.	``
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.authentication`	string (enum)	Yes	Authentication protocol for WPA3-Enterprise.	`peap`

Field	Type	Required	Description	Example
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.innerAuthentication`	string (enum)	No	Inner authentication method to be used within the outer authentication protocol.	`tls`
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.identity`	string	No	The identity (username) for WPA3-Enterprise authentication.	`test`
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.anonymousIdentity`	string	No	The anonymous identity for WPA3-Enterprise authentication.	`test`
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.password`	string	No	The password for WPA3-Enterprise authentication.	`test@123`
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.passphrase`	string	No	The passphrase for WPA3-Enterprise.	`test@123`
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.certificate`	array	No		`ap1wpa2e`
`wifiConfig.accessPoint.security.securityDetails.WPA3Enterprise.protocol`	string (enum)	No	The encryption protocol for WPA3-Enterprise.	`WPA3_Enterprise_CCMP`
`wifiConfig.accessPoint.security.securityDetails.OWEPublic`	object	No		``

Field	Type	Required	Description	Example
`wifiConfig.accessPoint.security.securityDetails.OWEPublic.authentication`	string	Yes	Authentication protocol for OWE-Public.	

Request Examples

Create WPA2 personal

```
{
  "command": "set_wifi",
  "requestId": "abc123",
  "wifiConfig": {
    "interfaceName": "wlan0",
    "enableInterface": true,
    "operation": "CREATE",
    "accessPoint": {
      "connect": false,
      "isPreferred": false,
      "autoConn": true,
      "essid": "TestAP1",
      "enableSecurity": true,
      "security": {
        "securityType": "WPA2Personal",
        "securityDetails": {
          "WPA2Personal": {
            "password": "test@123"
          }
        }
      }
    }
  }
}
```

Modify WPA2 personal

```
{
  "command": "set_wifi",
  "requestId": "abc123",
  "wifiConfig": {
    "interfaceName": "wlan0",
    "enableInterface": true,
    "operation": "MODIFY",
    "accessPoint": {
```

```
      "connect": false,
      "isPreferred": false,
      "autoConn": true,
      "essid": "TestAP1",
      "enableSecurity": true,
      "security": {
        "securityType": "WPA2Personal",
        "securityDetails": {
          "WPA2Personal": {
            "password": "test@12345678"
          }
        }
      }
    }
  }
}
```

Set preferred AP

```
{
  "command": "set_wifi",
  "requestId": "abc123",
  "wifiConfig": {
    "operation": "MODIFY",
    "accessPoint": {
      "isPreferred": true,
      "essid": "TestAP1"
    }
  }
}
```

Create WPA2 enterprise

```
{
  "command": "set_wifi",
  "requestId": "abc123",
  "wifiConfig": {
    "interfaceName": "wlan0",
    "enableInterface": true,
    "operation": "CREATE",
    "accessPoint": {
      "connect": false,
      "isPreferred": false,
      "autoConn": true,
      "essid": "TestAP2",
      "enableSecurity": true,
```

```
      "security": {
        "securityType": "WPA2Enterprise",
        "securityDetails": {
          "WPA2Enterprise": {
            "authentication": "tls",
            "innerAuthentication": "tls",
            "identity": "test",
            "anonymousIdentity": "test",
            "password": "test@123",
            "passphrase": "test@123",
            "certificate": [
              {
                "key": "ca_cert",
                "value": "gen_cert"
              },
              {
                "key": "client_key",
                "value": "gen_cl_key"
              },
              {
                "key": "client_cert",
                "value": "gen_cl_cert"
              }
            ],
            "protocol": "WPA2_Enterprise_CCMP"
          }
        }
      }
    }
  }
}
```

Response

Response Fields

Field	Type	Required	Description	Example
`command`	string	Yes		`set_wifi`
`requestId`	string	Yes		`abc123`
`apiVersion`	string (enum)	Yes		`V1.1`

Field	Type	Required	Description	Example
`response`	object	No	Status of the command execution — check `code` to confirm success (0) or identify the error.	

Field	Type	Required	Description	Example
`response.code`	integer	Yes	Response code indicating success or failure. • **0** — Success • **1** — Command payload accepted • **2** — Invalid payload • **3** — Not able to retrieve information • **4** — Firmware update in progress • **5** — Can't reboot, inventory in progress • **6** — Region is not configured • **7** — Interface is not available • **8** — Insufficient flash size • **9** — File not found • **10** — Configuration already exists • **11** — Inventory in progress • **12** — No radio operation in progress • **13** — Firmware update failed • **14** — Battery too low for firmware update • **15** — WIFI: SSID not found • **16** — WIFI: Cannot delete active SSID • **17** — WIFI: SSID missing	`0`

Field	Type	Required	Description	Example
			• **18** — WIFI: SSID already exists • **19** — WIFI: SSID count overflow • **20** — WiFi is not supported • **21** — Certificate not found • **22** — Advanced configuration not set • **23** — Invalid enum value • **24** — Max 32 prefilters limit exceeded • **25** — Max 3 publish topics exceeded • **26** — Max 1 subscribe topic exceeded • **27** — Invalid tenant ID length • **28** — Tag match pattern length exceeded	
`response.description`	string	Yes	Human-readable description of the response code (e.g. "Success", "Invalid payload").	`Success`

Response Examples

example1

```
{
  "command": "set_wifi",
  "requestId": "abc123",
  "apiVersion": "V1.1",
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```